

2013



Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen.
- Write your student number on each booklet.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- The mark allocated for each question is listed at the side of the question.
- Marks may be deducted for careless or poorly presented work.

Total Marks – 100

Section I – Pages 2 – 5

10 marks

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the sheet provided.

Section II – Pages 6 – 13

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.
- Begin each question in a new booklet.
- Show all necessary working in Questions 11 – 16.
- Templates for Q14(b) to be detached and placed in Q14 answer booklet.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I

10 marks

Marks

Attempt Questions 1 – 10

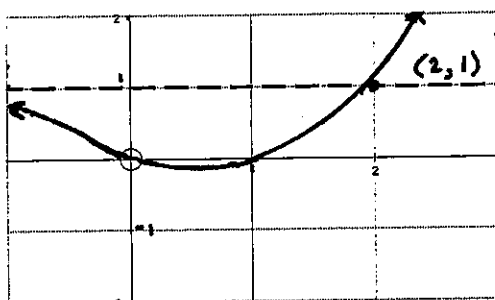
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. Let $z = 1 + 2i$ and $w = -2 + i$. What is the value of $\bar{z} \cdot w$

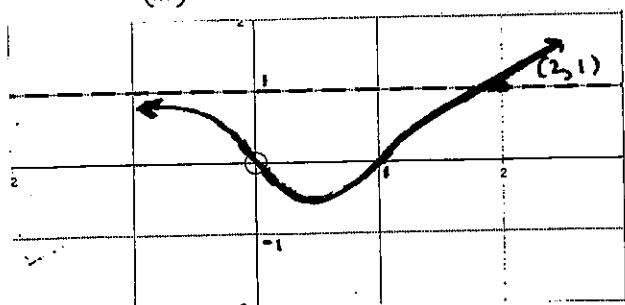
- (A) $5i$
- (B) $-4 + 5i$
- (C) $-3i$
- (D) $-4 - 3i$

2. The graph of $y = f(x)$ is shown below

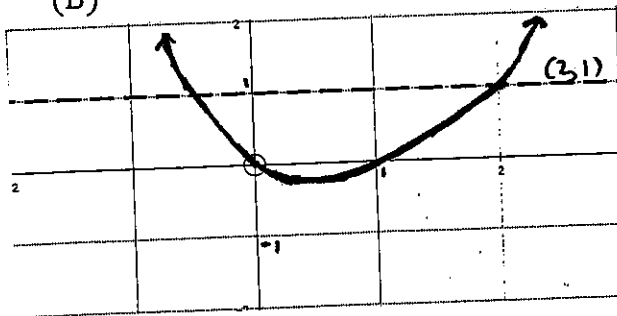


Which of the following is the graph of $y = \sqrt{f(x)}$?

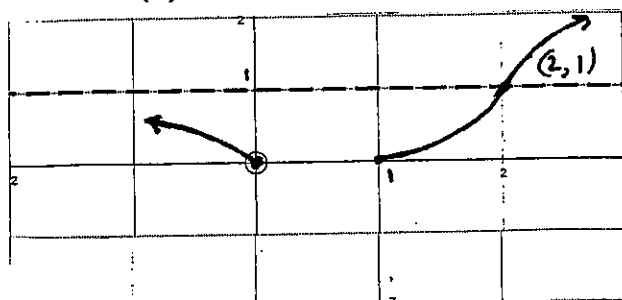
(A)



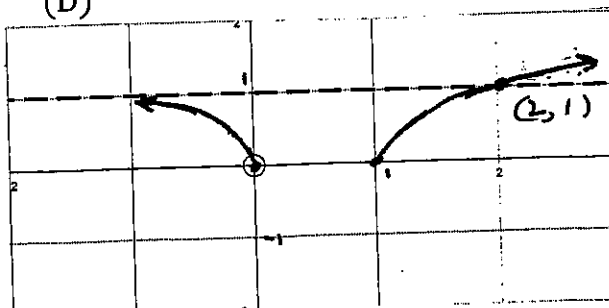
(B)



(C)

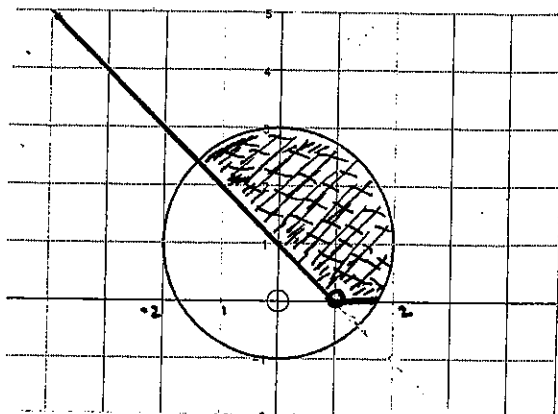


(D)



Section I (cont'd)

3. Consider the Argand diagram below.



Which inequality could define the shaded area?

- (A) $|z - i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$
- (B) $|z + i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{3\pi}{4}$
- (C) $|z - i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{\pi}{4}$
- (D) $|z + i| \leq 2$ and $0 \leq \arg(z - 1) \leq \frac{\pi}{4}$
4. The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the chord PQ subtends a right angle at $(0, 0)$. Which of the following is the correct expression?
- (A) $\tan \theta \cdot \tan \phi = -\frac{b^2}{a^2}$
- (B) $\tan \theta \cdot \tan \phi = -\frac{a^2}{b^2}$
- (C) $\tan \theta \cdot \tan \phi = \frac{b^2}{a^2}$
- (D) $\tan \theta \cdot \tan \phi = \frac{a^2}{b^2}$

Section I (cont'd)

5. Which of the following is an expression for $\int \frac{\sin x \cdot \cos x}{4 + \sin x} dx$

Use the substitution $u = 4 + \sin x$

- (A) $-4 \ln |4 + \sin x| + C$
(B) $4 \ln |4 + \sin x| + C$
(C) $-\sin x - 4 \ln |4 + \sin x| + C$
(D) $4 + \sin x - 4 \ln |4 + \sin x| + C$
6. The polynomial $P(x) = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$ has $x = 1$ as a root of multiplicity 3 and $x = i$ is a root. Which of the following expressions is a factorised form of $P(x)$ over the complex numbers?
- (A) $P(x) = (x + 1)^3 (x - 1)(x + 1)$
(B) $P(x) = (x - 1)^3 (x - 1)(x + 1)$
(C) $P(x) = (x + 1)^3 (x - i)(x + i)$
(D) $P(x) = (x - 1)^3 (x - i)(x + i)$
7. What is the eccentricity of the hyperbola

$$\frac{(x - 1)^2}{10} - \frac{(y + 1)^2}{4} = 1$$

- (A) $\frac{\sqrt{6}}{2}$ B. $\sqrt{\frac{7}{5}}$ C. $\frac{2}{\sqrt{6}}$ D. $\frac{\sqrt{14}}{2}$
8. The polynomial equation $x^3 - 3x^2 - x + 2 = 0$ has roots α, β and γ . Which of the following polynomial equations have roots $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$
- (A) $2x^3 - 2x^2 - 3x + 1 = 0$
(B) $2x^3 - x^2 - 3x + 1 = 0$
(C) $x^3 - 2x^2 - 3x + 1 = 0$
(D) $x^3 - x^2 - 3x + 1 = 0$

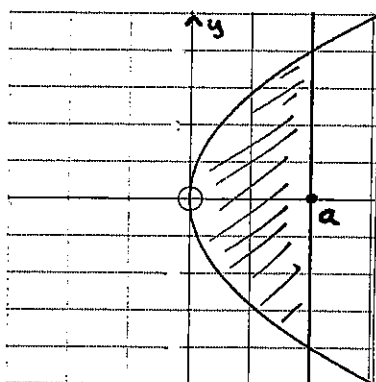
Section I (cont'd)

9. A particle of mass m is moving in a straight line under the action of a force

$$F = \frac{m}{x^3}(6 - 10x)$$

What of the following is an expression for its velocity in any position, if the particle starts from rest at $x = 1$?

- (A) $v = \pm \frac{1}{x} \sqrt{(-3 + 10x - 7x^2)}$
(B) $v = \pm x \sqrt{(-3 + 10x - 7x^2)}$
(C) $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$
(D) $v = \pm x \sqrt{2(-3 + 10x - 7x^2)}$
10. A solid is formed by rotating the region enclosed by the parabola $y^2 = 4ax$, its vertex $(0, 0)$ and the line $x = a$, about the y -axis.



What is the volume of this solid using the method of slicing.

- (A) $8\pi a^3$
(B) $\frac{16\pi a^3}{5}$
(C) $\frac{8\pi a^3}{5}$
(D) $4\pi a^3$

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Question 11 – Start A New Booklet – (15 marks)

Marks

- a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{3 + 5 \cos x} dx$ using the substitution $\tan\left(\frac{x}{2}\right) = t$ 3
- b) Let $z = 3 - 2i$ and $w = 1 + \sqrt{2}i$
- (i) Find $|z|$ 1
- (ii) Express $\frac{w}{z}$ in the form $a + bi$ where a and b are real numbers. 2
- c) Find $\int \frac{dx}{20 - 4x + x^2}$ 2
- d) (i) Write $z = 1 + i$ in modulus-argument form. 2
- (ii) Hence express z^{-4} in the form $x + yi$, where x and y are real. 2
- e) The area bounded by the curve $y = \frac{1}{x+1}$, the x -axis, the line $x = 2$ and the line $x = 8$, is rotated about the y -axis. 3
- Find the volume of the solid generated using the method of cylindrical shells.

Question 12 – Start A New Booklet – (15 marks)

Marks

a) Find $\int \frac{x}{(1-x)(1+x^2)} dx$

3

b) Let $f(x) = \frac{1-x}{x}$

On separate diagrams sketch the graph of the following functions. For each graph label any asymptotes and critical points.

(i) $y = |f(x)|$

2

(ii) $y = e^{f(x)}$

2

(iii) $y^2 = f(x)$

2

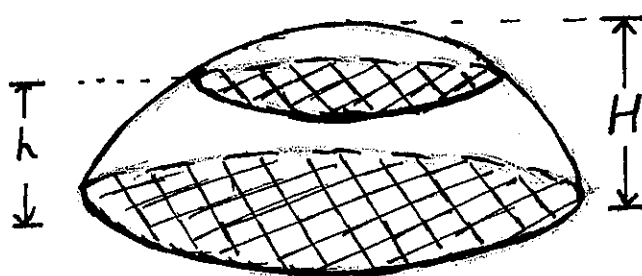
c) (i) Verify that $\int_0^a \sqrt{a^2 - x^2} dx = \frac{1}{4}\pi a^2$

1

(ii) Deduce that the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab

2

(iii)



The diagram shows a mound of height H . At height h above the horizontal base, the horizontal cross-section of the mound is an ellipse with equation

3

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \lambda^2 \text{ where } \lambda = 1 - \frac{h^2}{H^2}$$

(x, y are appropriate co-ordinates in the plane of the cross-section).

Show that the volume of the mound is $\frac{8\pi abH}{15}$

Question 13 – Start A New Booklet – (15 marks)

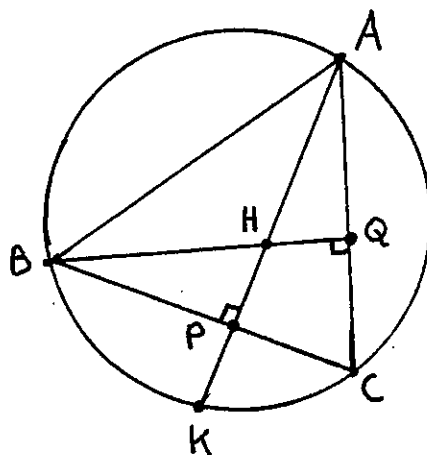
Marks

- a) On a school camp, one of the girls of mass M kg jumps vertically (feet first) from a rock ledge into a river below.

When she is falling at v m/s, she encounters air resistance equal to $\frac{Mv}{20}$ Newtons. She hits the water at a speed of V m/s. Let x be the displacement below the rock ledge at time t seconds after jumping.

- (i) Show that $\ddot{x} = g - \frac{v}{20}$, where g is the acceleration due to gravity. 1
- (ii) If it takes two seconds for her feet to hit the water, using $g = 10 \text{ m/s}^2$ show that $V = 200 \left(1 - e^{-\frac{1}{10}}\right)$ 3
- (iii) Find the height of the rock ledge above the water (to the nearest 0.1 metre) 3

b)



The altitudes AP and BQ of the acute triangle ABC intersect at H . 3

AP produced cuts the circle at K .

Prove that $HP = PK$

Question 13 (cont'd)

Marks

- c) Show that the equation of the tangent to the ellipse

1

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b > 0)$$

at the point $P(a \cos \theta, b \sin \theta)$ is $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$

- (i) The tangent at P meets the x -axis at S and the y -axis at T .

Find the area of ΔOST .

2

- (ii) If A is the point $(a, 0)$ and B is the point $(0, b)$, the area of ΔAPB is $\frac{1}{2} ab(\cos \theta + \sin \theta - 1)$ [do not prove this]

Prove that, as θ varies in the interval $0 < \theta < \frac{\pi}{2}$, the area of ΔAPB is a maximum when the tangent to the ellipse is parallel to AB .

2

Question 14 – Start A New Booklet – (15 marks)

Marks

- a) On an Argand diagram, sketch the locus of the points z such that

2

$$\arg \left\{ \frac{(z-1)}{(z+1)} \right\} = \frac{\pi}{2}$$

- b) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are two points on the rectangular hyperbola $xy = c^2$.

Given that the equation of the tangent at P is

$$x + p^2y = 2cp.$$

- (i) The tangents at P and Q meet in T . Find the co-ordinates of T in terms of c, p and q .

2

- (ii) If T lies on the hyperbola $xy = k^2$ for all positions of P and Q , prove that

$$\frac{pq}{(p+q)^2} = \frac{k^2}{4c^2}$$

1

- c) (i) Given $z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1)$ let w be a solution to $z^5 + 1 = 0$ where $w \neq -1$. Prove that $1 + w^2 + w^4 = w + w^3$

1

- (ii) Hence show that

3

$$\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$$

Question 14 (cont'd)

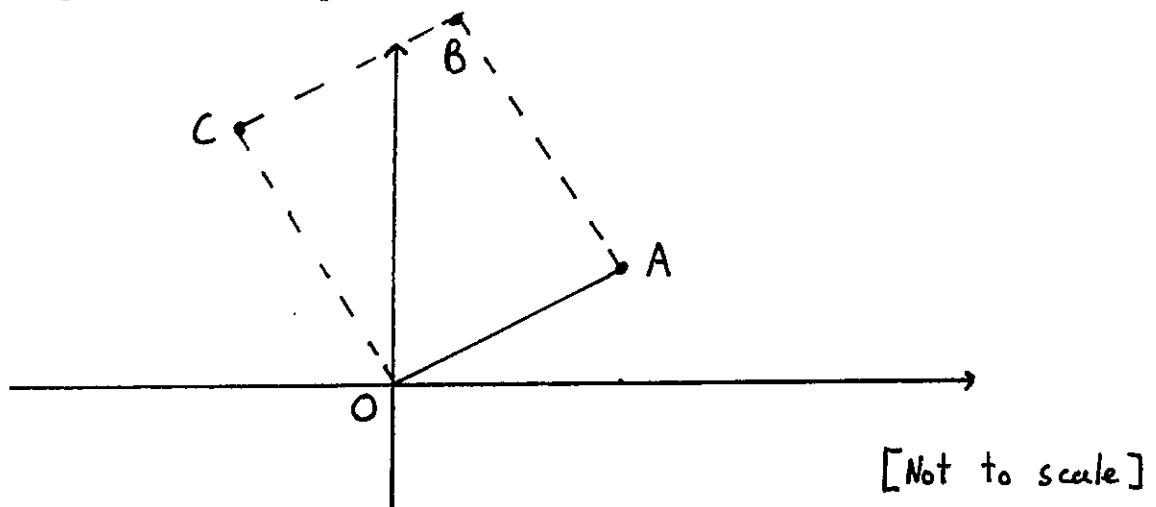
Marks

- d) (i) If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ where $n \geq 1$, use integration by parts to prove that 3

$$I_n = \frac{n-1}{n} I_{n-2}$$

- (ii) Hence show that $I_5 = \frac{8}{15}$ 1

- e) $OABC$ is a rectangle in an Argand diagram where O is the Origin and point A corresponds to the complex number $2 + i$



Given that the length of the rectangle is twice its breadth and OA is one of the shorter sides, find the complex number representing C .

2

Question 15 – Start A New Booklet – (15 marks)

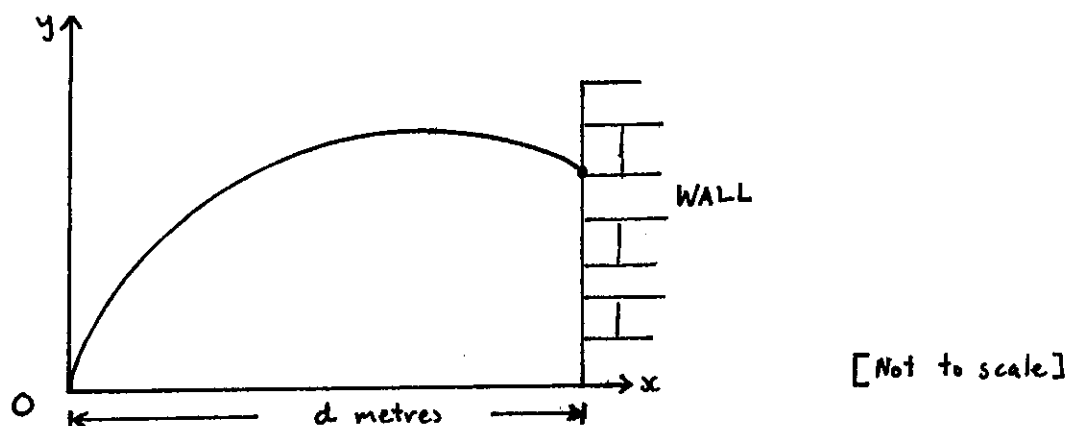
Marks

- a) Suppose α, β and γ are the roots of the polynomial equation

$$x^3 + x + 12 = 0$$

- (i) Find $\alpha^2 + \beta^2 + \gamma^2$ 2
- (ii) Hence explain why only one of the roots is real. 2
- (iii) Let the real root be denoted by α . Prove that $-3 < \alpha < -2$ 1
- (iv) Hence prove that the modulus of each of the other roots lies between 2 and $\sqrt{6}$ 3

b)



In the diagram above, the wall of a building stands on level ground, d metres from a fire hose located at O . If water leaves the hose with velocity $V \text{ ms}^{-1}$ at an angle θ to the ground:

Given $V > \sqrt{gd}$, (g is acceleration in the vertical plane due to gravity) show that the particle will strike the wall above ground level provided that $\beta < \theta < \frac{\pi}{2} - \beta$ where $\beta = \frac{1}{2} \sin^{-1}\left(\frac{gd}{V^2}\right)$

You may assume that the range on the horizontal plane from the point of projection is

$$\frac{V^2 \sin 2\theta}{g}$$

- c) Find $\int \frac{dx}{x^{\frac{1}{2}} + x^{\frac{1}{3}}}$ 3

[Hint: • choose an appropriate substitution]

$$\bullet \frac{x^3}{x+1} = x^2 - x + 1 - \frac{1}{x+1}$$

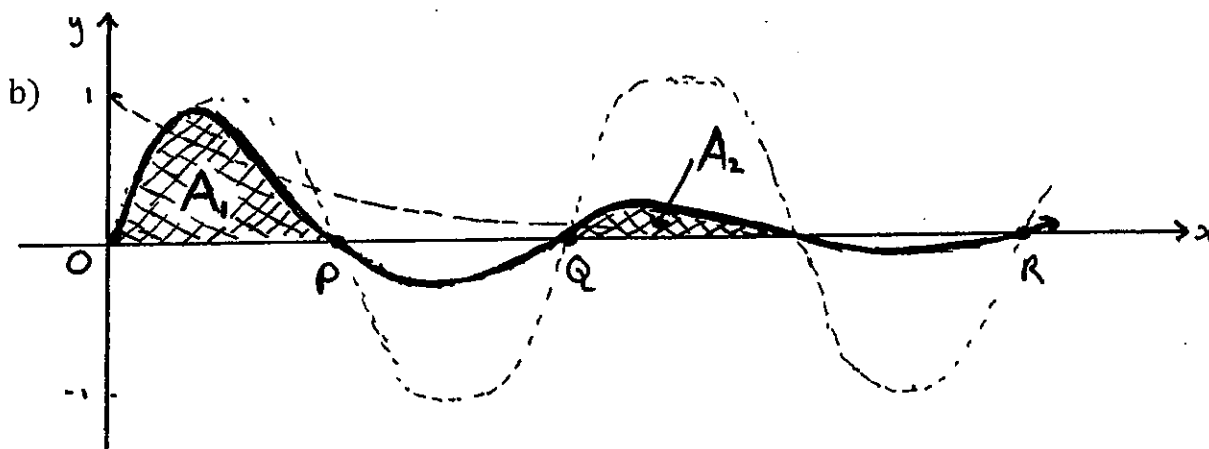
Question 16 – Start A New Booklet – (15 marks)

Marks

- a) A polynomial of degree n is given by $P(x) = x^n + ax - b$. It is given that the polynomial has a double root at $x = \alpha$.

(i) Find the derived polynomial $P'(x)$ and show that $\alpha^{n-1} = -\frac{a}{n}$ 2

(ii) Show that $\left(\frac{a}{n}\right)^n + \left(\frac{b}{n-1}\right)^{n-1} = 0$ 4



The diagram shows a sketch of part of the curve $y = f(x)$ with equation

$$y = e^{-x} \cdot \sin x, \quad x \geq 0$$

- (i) Find the coordinates of the points P , Q and R where $y = f(x)$ cuts the x -axis. 1

- (ii) Integrating by parts gives 3

$$\int e^{-x} \cdot \sin x \, dx = -\frac{1}{2} e^{-x} (\sin x + \cos x) + C$$

[you do not have to prove this]

If the terms $A_1, A_2, \dots, A_n$, represent areas between $y = f(x)$ and the x -axis for successive portions of $y = f(x)$ where y is positive. (The areas represented by A_1 , and A_2 , are shown as the shaded regions in the diagram).

Show that $A_n = \frac{1}{2} (e^{(1-2n)\pi} + e^{(2-2n)\pi})$

- (iii) Show that $A_1 + A_2 + A_3 + \dots$ is a geometric series and that 3

$$S_{\infty} = \frac{e^{\pi}}{2(e^{\pi} - 1)}$$

- (iv) Given that $\int_0^{\infty} e^{-x} \cdot \sin x \, dx = \frac{1}{2}$, find the exact value of $\int_0^{\infty} |e^{-x} \cdot \sin x| \, dx$ 2

Student Number: _____ Teacher: _____

Year 12 Mathematics Extension 2 Trial HSC Examination 2013

Section I

Multiple-choice Answer Sheet – Questions 1 – 10

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
correct

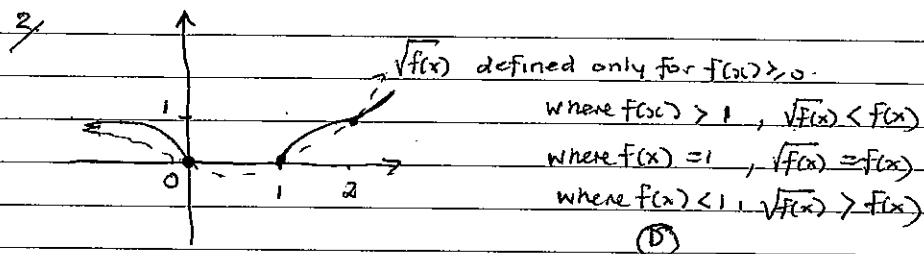
-
- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

(1)

2013 EXT2 TRIAL HSC

SECTION I

1/ $z = 1+2i$ $w = -2+i$
 $\bar{z} = 1-2i$ $\bar{z} \cdot w = (1-2i)(-2+i)$
 $= -2 + 5i - 2i^2$
 $= 5i$ (A)



3/ $x^2 + (y-1)^2 \leq 4$
 $|z-i| \leq 2$ and $0 \leq \arg(z-i) \leq \frac{3\pi}{4}$ (A)

4/ $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$
 lie on $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ chord PQ subtends a right angle at $(0,0)$

$$m_1 \times m_2 = -1$$

$$\frac{b \sin \theta}{a \cos \theta} \times \frac{b \sin \phi}{a \cos \phi} = -1$$

$$\frac{b^2 \tan \theta \times \tan \phi}{a^2} = -1 \quad \therefore \tan \theta \tan \phi = -\frac{a^2}{b^2} \quad (B)$$

5/ $\int \frac{\sin x \cdot \cos x}{4 + \sin x} dx$
 $u = 4 + \sin x \quad \therefore \sin x = u - 4$
 $\frac{du}{dx} = \cos x \quad \therefore du = \cos x dx$
 $\int \frac{(u-4) du}{u}$
 $= \int 1 - \frac{4}{u} du$
 $= u - 4 \ln u + c$
 $= 4 + \sin x - 4 \ln |4 + \sin x| + c$ (D)

(2)

2013 EXT2 TRIAL HSC

A

6/ $P(x) = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$
 $= (x-1)^3(x-i)(x+i)$
 $(-1)^3 \times (-i) \times (i) = -1 \times -i^2$
 $= -1$ (D)

7/ $\frac{(x-1)^2}{10} - \frac{(y+1)^2}{4} = 1$ hyperbola $e = \sqrt{1 + \frac{b^2}{a^2}}$ $b^2 = 4$ $a^2 = 10$
 $= \sqrt{1 + \frac{4}{10}}$
 $= \sqrt{\frac{14}{5}}$ (B)

8/ $x^3 - 3x^2 - x + 2 = 0$

$x = \frac{1}{x}$, $x = \frac{1}{x}$ $(\frac{1}{x})^3 - 3(\frac{1}{x})^2 - (\frac{1}{x}) + 2 = 0$
 $\frac{1}{x^3} - \frac{3}{x^2} - \frac{1}{x} + 2 = 0$
 $1 - 3x - x^2 + 2x^3 = 0$ (B)

B

9/ $F = \frac{m}{x^3} (6 - 10x)$ $F = mg$
 $a = 6x^{-3} - 10x^{-2}$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 6x^{-3} - 10x^{-2}$$

$$\frac{1}{2} v^2 = \int 6x^{-3} - 10x^{-2} dx$$

$$\frac{1}{2} v^2 = -3x^{-2} + 10x^{-1} + c$$

when $x=1$, $v=0$ $\therefore c = -7$

$$\frac{1}{2} v^2 = -3x^{-2} + 10x^{-1} - 7$$

$$v^2 = 2(-3x^{-2} + 10x^{-1} - 7)$$

$$= \frac{2}{x^2} (-3 + 10x - 7x^2)$$

$$v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$$
 (C)

QUESTION 11

(a) let $t = \tan\left(\frac{x}{2}\right)$

then $\tan^{-1}(t) = \frac{x}{2}$

$\therefore \frac{2 dt}{1+t^2} = dx$

When $x=0$, $t=0$

$x = \frac{\pi}{2}$, $t=1$

Gives on substitution

$$\int_0^1 \frac{1}{3 + 5 \frac{(1-t^2)}{(1+t^2)}} \cdot \frac{2 dt}{(1+t^2)}$$

$$= \int_0^1 \frac{2 dt}{3(1+t^2) + 5(1-t^2)}$$

$$* = \int_0^1 \frac{dt}{4-t^2}$$

$$= \int_0^1 \frac{\frac{1}{4}}{(2-t)} + \frac{\frac{1}{4}}{(2+t)} dt$$

$$= \frac{1}{4} \left[-\ln(2-t) + \ln(2+t) \right]_0^1$$

$$= \frac{1}{4} \left[\ln \left[\frac{2+t}{2-t} \right] \right]_0^1$$

$$= \frac{1}{4} (\ln 3 - \ln 1)$$

$$= \frac{1}{4} \ln 3$$

* Now: $\frac{1}{4-t^2} = \frac{A}{2-t} + \frac{B}{2+t}$

So $1 = 2(A+B) + t(A-B)$

$A-B=0 \quad \dots (I)$

$A+B=\frac{1}{2} \quad \dots (II)$

$(I)+(II) \quad 2A = \frac{1}{2}$

$A = \frac{1}{4}, B = \frac{1}{4}$

(b) (i) $|z| = \sqrt{3^2 + (-2)^2}$

$= \sqrt{13}$

(ii) $\frac{w}{z} = \frac{1 + \sqrt{2}i}{3 - 2i} \times \left(\frac{3 + 2i}{3 + 2i} \right)$

$$= \frac{3 + 2i + 3\sqrt{2}i - 2\sqrt{2}}{3^2 + 4}$$

$$= \frac{3-2\sqrt{2}}{13} + \frac{2+3\sqrt{2}}{13} i$$

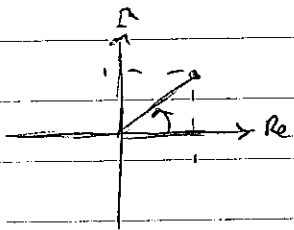
$$(c) \int \frac{dx}{2x^2 - 4x + x^2} = \int \frac{dx}{x^2 - 4x + 4 + 16}$$

$$= \int \frac{dx}{(x-2)^2 + 4^2}$$

$$= \frac{1}{4} \tan^{-1} \left(\frac{x-2}{4} \right) + C$$

$$(d) (i) \text{ Mod } |z| = \sqrt{1^2 + 1^2} \text{ and Arg } \arg(z) = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

$$= \sqrt{2}$$



$$\therefore z = r \operatorname{cis} \theta$$

$$= \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

$$= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$(ii) \text{ Then } z^{-4} = \left(\sqrt{2} \cdot \operatorname{cis} \frac{\pi}{4} \right)^{-4}$$

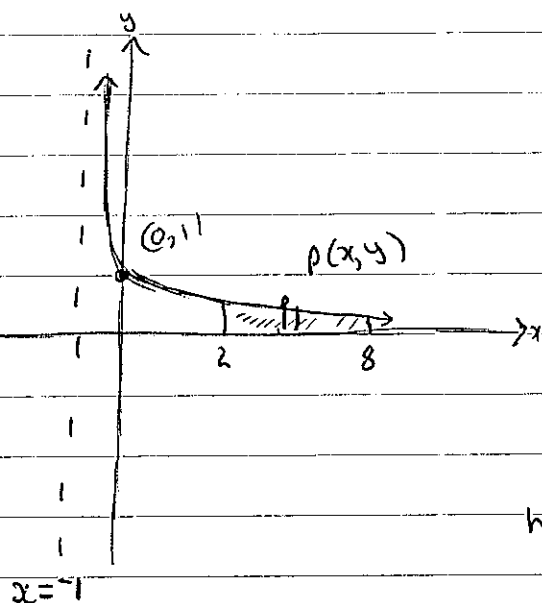
$$= 2^{-2} \cdot \operatorname{cis} \left(-4 \times \frac{\pi}{4} \right) \text{ (by De Moivre's Theorem)}$$

$$= \frac{1}{4} \left(\cos(-\pi) + i \sin(-\pi) \right)$$

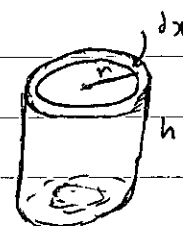
$$= \frac{1}{4} (-1 - 0 \cdot i)$$

$$= -\frac{1}{4} \quad (\text{or } -\frac{1}{4} + 0 \cdot i)$$

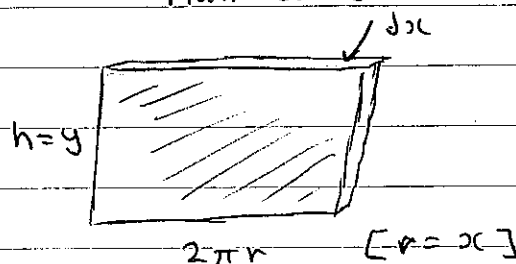
(e)



"Typical" shell



"Flattened out"



$$\begin{aligned}
 \text{So } dV &= A \cdot dx \\
 &= 2\pi x \cdot y \cdot dx \\
 &= 2\pi x \cdot \frac{1}{x+1} dx
 \end{aligned}$$

$$\text{Now } V = \lim_{dx \rightarrow 0} \sum_{x=2}^8 2\pi \cdot \frac{x}{x+1} dx$$

$$= 2\pi \int_2^8 \frac{x}{x+1} dx$$

$$= 2\pi \int_2^8 \left(\frac{x+1}{x+1} - \frac{1}{x+1} \right) dx$$

$$= 2\pi \left[x - \ln|(x+1)| \right]_2^8$$

$$= 2\pi \left[(8 - \ln(9)) - (2 - \ln(3)) \right]$$

$$= 2\pi \left(6 - \ln\left(\frac{9}{3}\right) \right)$$

$$= 2\pi (6 - \ln 3) u^2$$

QUESTION 12

(a) Now, partial fractions gives

$$\frac{x}{(1-x)(1+x^2)} = \frac{A}{(1-x)} + \frac{Bx+C}{(1+x^2)}$$

$$x = A + Ax^2 + Bx - Bx^2 + C - Cx$$

$$= (A-B)x^2 + (B-C)x + (A+C)$$

Gives: $A-B=0$ -- (I)

$B-C=1$ -- (II)

$A+C=0$ -- (III)

$\therefore A=B$

So $A-C=1$

+ $A+C=0$

$2A=1$

$A=\frac{1}{2}$

Have $A=\frac{1}{2}, B=\frac{1}{2}, C=-\frac{1}{2}$

Integral

$$* \int \frac{x}{(1-x)(1+x^2)} dx$$

$$= \frac{1}{2} \int \left(\frac{1}{1-x} + \frac{x-1}{1+x^2} \right) dx$$

$$= \frac{1}{2} \int \left(\frac{1}{1-x} + \frac{x}{1+x^2} - \frac{1}{1+x^2} \right) dx$$

$$= \frac{1}{2} \left[-\ln|1-x| + \frac{1}{2} \ln(1+x^2) - \tan^{-1}x \right] + C$$

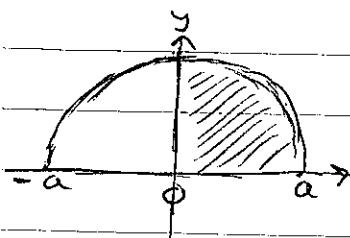
* $x < 1$

$$= \frac{1}{2} \left[\ln \sqrt{(1+x^2)^{\frac{1}{2}}} - \frac{1}{2} \tan^{-1}x \right] + C$$

(b) See attached sheet.

(c) (i) $\int_0^a \sqrt{a^2-x^2} dx$

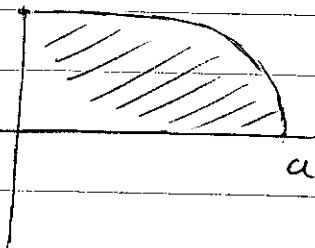
describes the area of a quadrant of the circle radius 'a' units, centre (0,0)



So, $A = \frac{1}{4} \pi r^2$

Gives $\int_0^a \sqrt{a^2-x^2} dx = \frac{1}{4} \pi a^2$

(ii) b



Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

gives $y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$

Area of one quadrant is

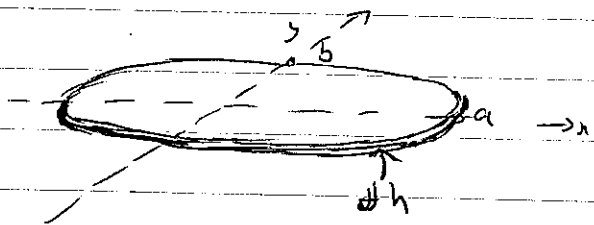
$$= \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$$

$$= \frac{b}{a} \cdot \frac{1}{4} \pi a^2 \quad [\text{from (i)}]$$

$$= \frac{1}{4} \pi a b.$$

$\therefore$ Area of four quadrants $4 \times \frac{1}{4} \pi a b$
 $= \pi a b.$

(iii) "Typical" cross section is an ellipse in the x - y plane



Given $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \lambda^2$

then $\frac{x^2}{\lambda^2 a^2} + \frac{y^2}{\lambda^2 b^2} = 1$

and from part (ii) Area = $\pi (a\lambda) (b\lambda)$

$$= \pi a b \lambda^2$$

So

$$\delta V = A \cdot \delta h$$

$$V = \lim_{\delta h \rightarrow 0} \sum_0^H \pi \cdot a b \lambda^2 \cdot dh$$

$$= \int_0^H \pi a b \lambda^2 dh$$

$$= \pi a b \int_0^H \left[1 - \frac{h^2}{H^2} \right]^2 dh$$

$$= \pi a b \int_0^H \left(1 - 2 \frac{h^2}{H^2} + \frac{h^4}{H^4} \right) dh$$

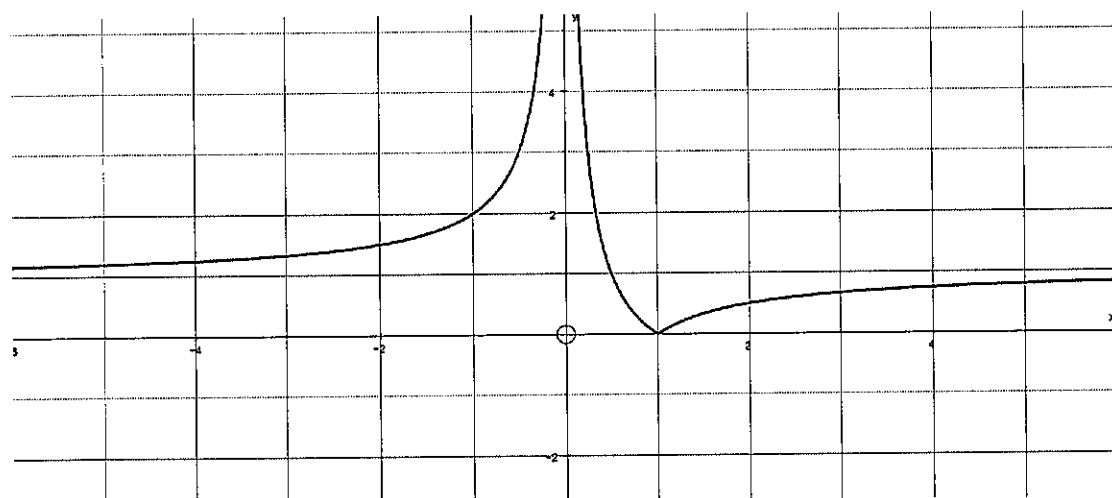
$$= \pi a b \left[h - \frac{2}{3} \cdot \frac{h^3}{H^2} + \frac{1}{5} \cdot \frac{h^5}{H^4} \right]_0^H$$

$$= \pi a b \left(H - \frac{2}{3} \cdot H + \frac{1}{5} \cdot H \right)$$

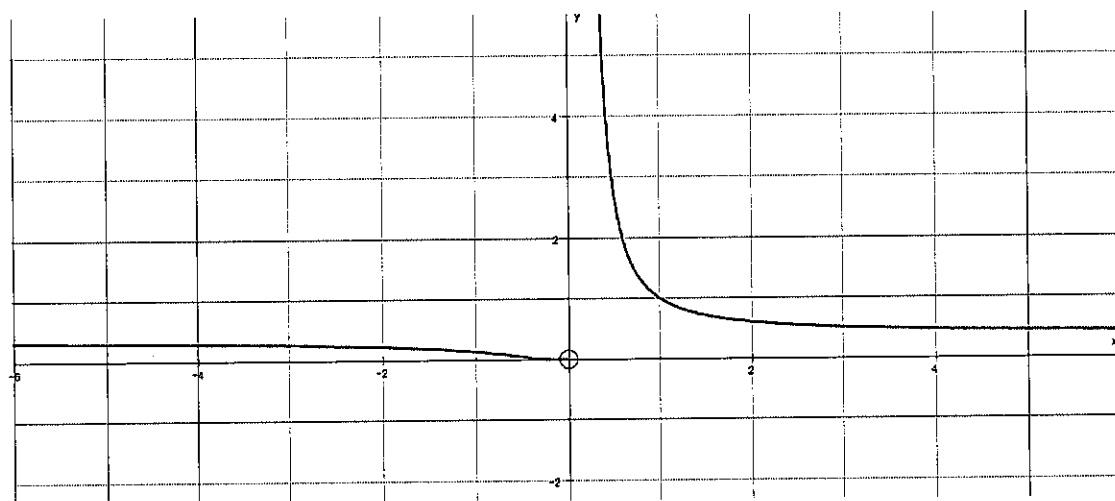
$$V = \frac{8 \pi a b H}{15} \omega^3$$

Graphs

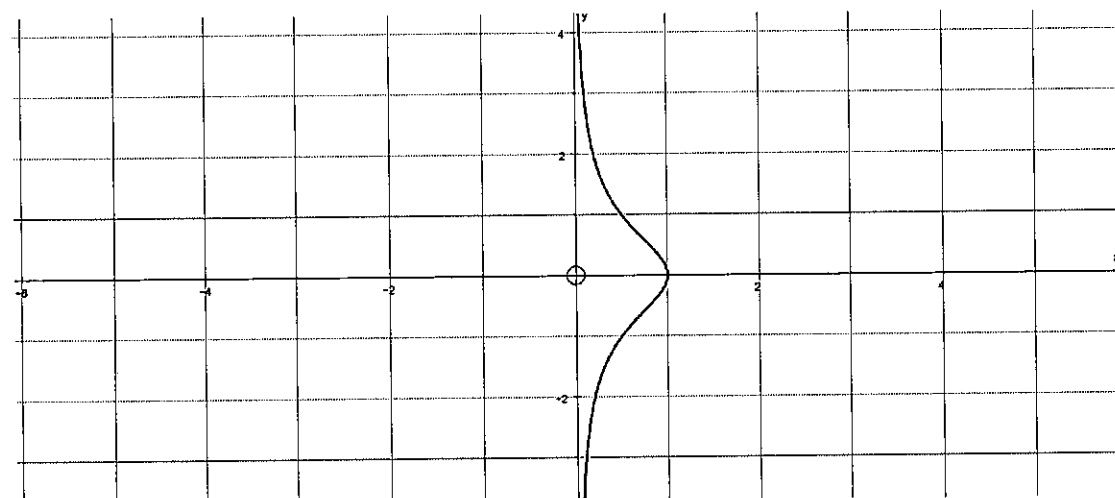
(i). $y = \left| \frac{1-x}{x} \right|$



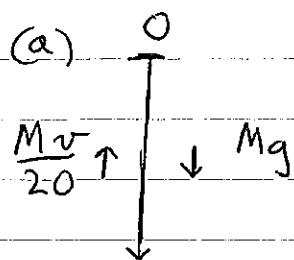
(ii). $y = e^{\frac{1-x}{x}}$



(iii). $y^2 = \frac{1-x}{x}$



QUESTION 13



Direction diagram

$$(i) \quad F = Mg - \frac{Mv}{20}$$

$$\therefore M\ddot{x} = Mg - \frac{Mv}{20}$$

$$\ddot{x} = g - \frac{v}{20}$$

$$(ii) \quad \ddot{x} = \frac{dv}{dt} = \frac{20g - v}{20}$$

$$\frac{dt}{dv} = \frac{20}{20g - v}$$

Thus $\frac{dt}{20} = \frac{dv}{20g - v}$

$$\frac{t}{20} = -\ln(20g - v) + C$$

at $t=0, v=0$ so $C = \ln(20g)$

Then $\frac{t}{20} = -\ln\left[\frac{20g - v}{20g}\right]$

$$e^{-\frac{t}{20}} = \frac{20g - v}{20g}$$

$$20ge^{-\frac{t}{20}} = 20g - v$$

* $v = 20g(1 - e^{-\frac{t}{20}})$

at $t=2$ gives $V = 20g(1 - e^{-\frac{1}{10}})$
 $v = V$

(ii) Also $\ddot{x} = v \frac{dv}{dx} = \frac{20g-v}{20}$

$$\frac{dv}{dx} = \frac{20g-v}{20}$$

$$= \frac{20g-v}{20v}$$

So $\frac{dx}{dv} = \frac{20v}{20g-v}$

$$dx = \frac{20v}{20g-v} \cdot dv$$

$$\therefore x = -20 \int \frac{20g-v-20g}{20g-v} dv$$

$$= -20 \int \frac{20g-v}{20g-v} - \frac{20g}{20g-v} dv$$

Crosses $x = -20 \int 1 - \frac{20g}{20g-v} dv$

$$= -20 [v + 20g \ln(20g-v)] + C$$

at $x=0, v=0$ So $C = 20[20g \ln 20g]$

$$\therefore x = -20 \left[v + 20g \ln \left[\frac{20g-v}{20g} \right] \right]$$

at $t=2$, $x=h$ and $v = V = 20g(1 - e^{-\frac{1}{10}})$

We have $h = -20 \left[20g(1 - e^{-\frac{1}{10}}) + 20g \ln(e^{-\frac{1}{10}}) \right]$

($g=10$)

$$= -20 [200(1 - e^{-\frac{1}{10}}) - 20]$$

≈ 19.3 metres is height rock ledge.

het $\hat{B}_{KA} = 0$

Then $A \hat{C} B = 0$

[angles at the circumference standing on the same arc are equal]

and $\hat{P}_{AC} = 90^\circ - \theta$

[angle sum of $\triangle APC$]

$\therefore \angle B\hat{C} = 90^\circ - \theta$ [Angles at circumference standing on the same arc are equal]

gives $\angle \hat{B}C = 90^\circ - \theta$ [angle sum of $\triangle QBC$]

Then,

$$\angle B \hat{P} K = 90^\circ = \angle B \hat{P} H \quad [\text{given in data}]$$
$$\angle K \hat{B} P = 90^\circ - \theta = \angle H \hat{B} C \quad [\text{from above}]$$

PB is common

$$\triangle KBP \equiv \triangle HBP \text{ by congruency test}$$

$$AAS$$

In congruent Δ 's corresponding sides are equal, $HP = PK$

(c) Gradient function $\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$

$$\therefore \frac{dy}{dx} = - \frac{b^2 x}{a^2 y}$$

at $P(a \cos \theta, b \sin \theta)$

$$m = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta}$$

$$= \frac{-b \cos \theta}{a \sin \theta}$$

So Equation of tangent

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\frac{y \sin \theta}{b} - \sin^2 \theta = \frac{-x \cos \theta}{a} + \cos^2 \theta$$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = \sin^2 \theta + \cos^2 \theta$$

$$= 1$$

(i) Let $y = 0$, then $\frac{x \cos \theta}{a} = 1$

$$x = \frac{a}{\cos \theta}$$

$$S\left(\frac{a}{\cos \theta}, 0\right)$$

and $x = 0$, then $\frac{y \sin \theta}{b} = 1$

$$y = \frac{b}{\sin \theta}$$

$$T\left(0, \frac{b}{\sin \theta}\right)$$

$$\therefore \text{Area of } \triangle OST = \frac{1}{2} \cdot \frac{a}{\cos \theta} \cdot \frac{b}{\sin \theta}$$

$$= \frac{ab}{\sin 2\theta}.$$

(ii) Let area be L for $\triangle APB$

then $\frac{dL}{d\theta} = \frac{1}{2} ab [-\sin \theta + \cos \theta]$

and $\frac{d^2L}{d\theta^2} = \frac{1}{2} ab [-\cos \theta - \sin \theta]$

$$= -\frac{1}{2} ab [\cos \theta + \sin \theta]$$

if $ab > 0$

and

$$0 \leq \theta \leq \pi$$

$$\cos \theta + \sin \theta \geq 0$$

$\therefore$ concave down.

$$\text{Let } \frac{dL}{d\theta} = 0, \Rightarrow -\sin \theta + \cos \theta = 0 \quad [ab \neq 0]$$

$$\sin \theta = \cos \theta$$

$$\tan \theta = 1 \quad [\theta \neq \pi]$$

$$\theta = \frac{\pi}{4}$$

Maximum Value at $\theta = \frac{\pi}{4}$

$$\text{So } P\left(a \cos \frac{\pi}{4}, b \sin \frac{\pi}{4}\right) = P\left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}}\right)$$

Gradient of tangent

$$m = -\frac{b^2 \frac{a}{\sqrt{2}}}{a^2 \frac{b}{\sqrt{2}}}$$

$$= -\frac{b}{a}$$

Gradient of AB

$$m_{AB} = \frac{b-0}{0-a}$$

$$= -\frac{b}{a}$$

Gradient of tangent equals gradient of AB, lines parallel for maximum area.

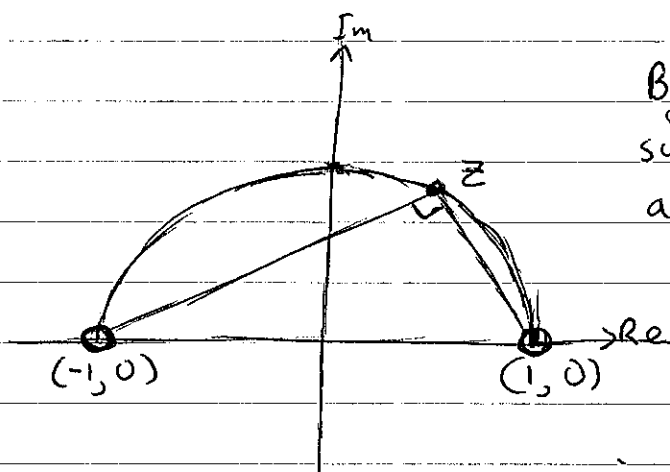
QUESTION 14

$$(a) \arg \left(\frac{z-1}{z+1} \right) = \arg(z-1) - \arg(z+1)$$

$$\text{So } \arg(z-1) - \arg(z+1) = \frac{\pi}{2} \quad (\text{given})$$

$$\therefore \arg(z-1) = \frac{\pi}{2} + \arg(z+1)$$

$$\begin{aligned} \text{Let } \arg(z-1) &= \theta \\ \arg(z+1) &= \beta \end{aligned} \Rightarrow \theta = \frac{\pi}{2} + \beta$$



By exterior angle equals sum of opposite interior angles $\Rightarrow z$ moves on a semi-circle [radius 1] above the x -axis.

[* Angle in a semi-circle is a right angle]

$$(b) (i) \text{ Data: tangent at } P \Rightarrow x + p^2 y = 2cp \quad \dots (I)$$

$$\text{by symmetry at } Q \Rightarrow x + q^2 y = 2cq \quad \dots (II)$$

$$(I) - (II) \Rightarrow (p^2 - q^2)y = 2c(p - q)$$

$$y = \frac{2c}{p+q}$$

$$\text{Substitute } x + p^2 \cdot \frac{2c}{p+q} = 2cp$$

$$x = 2cp \left(1 - \frac{p}{p+q} \right)$$

$$x = \frac{2cp}{p+q} (p+q - p)$$

$$x = \frac{2cpq}{p+q}$$

$$\therefore T \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$$

(ii) By substitution
in $xy = k^2$ $\frac{pq}{(p+q)^2} \Rightarrow \frac{2cpq}{p+q} \times \frac{2c}{p+q} = k^2$

$$\frac{4c^2 pq}{(p+q)^2} = k^2$$

$$\frac{pq}{(p+q)^2} = \frac{k^2}{4c^2}$$

(c) (i) If ω is a solution and $\omega \neq 1$
then $\omega + 1 \neq 0$

Given $\omega^5 + 1 = 0$
then $(\omega + 1)(\omega^4 - \omega^3 + \omega^2 - \omega + 1) = 0$

$\omega + 1 \neq 0 \quad \therefore \omega^4 - \omega^3 + \omega^2 - \omega + 1 = 0$
 $\omega^4 + \omega^2 + 1 = \omega^3 + \omega$

(ii) let $z = r \operatorname{cis} \theta$

now $|z| = r, \quad |z^5| = |z|^5 = 1$

So $r = 1$

gives $z = \cos \theta + i \sin \theta$

$$z^5 = \cos 5\theta + i \sin 5\theta \quad [\text{De Moivre}]$$

Equating $\cos 5\theta + i \sin 5\theta = -1 + 0.i$

then $\cos 5\theta = -1$ and $\sin 5\theta = 0$

$$\therefore 5\theta = \pm\pi, \pm 3\pi, \pm 5\pi \dots$$

$$\theta = \pm \frac{\pi}{5}, \pm \frac{3\pi}{5}, \pm \pi, \pm \frac{7\pi}{5}, \pm \frac{9\pi}{5} \dots$$

(i) $\theta = \pi$, $\cos \pi + i \sin \pi = -1$

(ii) $\theta = \frac{\pi}{5}$, $\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} = \omega$

(iii) $\theta = -\frac{\pi}{5}$, $\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} = \bar{\omega} = \cos \frac{\pi}{5} - i \sin \frac{\pi}{5}$

(iv) $\theta = \frac{3\pi}{5}$, $\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} = \omega^3$

(v) $\theta = -\frac{3\pi}{5}$, $\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} = \bar{\omega}^3 = \cos \frac{3\pi}{5} - i \sin \frac{3\pi}{5}$

Sum of roots for $z^4 - z^3 + z^2 - z + 1 = 0$

i) $\omega + \bar{\omega} + \omega^3 + \bar{\omega}^3 = 1$

$$2 \left(\cos \frac{\pi}{5} \right) + 2 \cos \left(\frac{3\pi}{5} \right) = 1$$

$$\cos \left(\frac{\pi}{5} \right) + \cos \left(\frac{3\pi}{5} \right) = \frac{1}{2}$$

(d) (i) Given $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \cdot dx$ $n \geq 1$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cdot \sin^2 x \cdot dx$$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \cos^2 x) dx$$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cdot dx - \int_0^{\frac{\pi}{2}} \cos^2 x \cdot \sin^{n-2} x \cdot dx$$

$$\begin{aligned}
 I_n &= I_{n-2} - \int_0^{\frac{\pi}{2}} \underbrace{\cos x}_u \left(\underbrace{\cos x \cdot \sin^{n-2} x}_{dv} \right) dx \\
 &= I_{n-2} - \left[\left(\cos x \cdot \frac{1}{n-1} \sin^{n-1} x \right) \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} -\sin x \cdot \frac{1}{n-1} \sin^{n-1} x dx \right] \\
 &= I_{n-2} - \left[0 + \frac{1}{n-1} \int_0^{\frac{\pi}{2}} \sin^n x dx \right]
 \end{aligned}$$

Now $I_n = I_{n-2} - \frac{1}{n-1} I_n$

$$I_n \left(1 + \frac{1}{n-1} \right) = I_{n-2} \quad \left[1 + \frac{1}{n-1} = \frac{n-1+1}{n-1} \right]$$

$$\frac{n}{n-1} I_n = I_{n-2} \quad = \frac{n}{n-1} \quad \Bigg]$$

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$

(ii) From (i) $I_5 = \frac{4}{5} I_3$

$$I_3 = \frac{2}{3} I_1$$

$$I_1 = \int_0^{\frac{\pi}{2}} \sin x dx$$

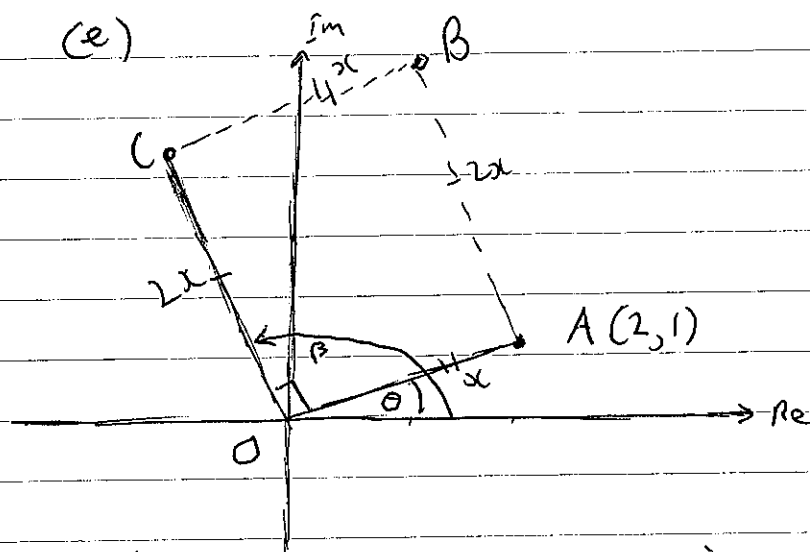
$$= \left[-\cos x \right]_0^{\frac{\pi}{2}}$$

$$= -(0 - 1)$$

$$= 1$$

Then $I_5 = \frac{4}{5} \left(\frac{2}{3} \times 1 \right)$

$$= \frac{8}{15}$$



For A, $|2+i| = \sqrt{5}$

$\therefore OA = \sqrt{5}$ and $OC = 2\sqrt{5}$

Now, OC is given by rotating OA through $\frac{\pi}{2}$ and doubling the length

$$\begin{aligned} \underline{\text{So}} \quad \beta &= \arg(OC) = \arg(2+i) + \frac{\pi}{2} \\ &= \arg(2+i) + \arg(ki) \\ &= \arg[k(-1+2i)] \end{aligned}$$

$$\begin{aligned} \text{Where } |k(-1+2i)| &= \sqrt{k^2 \cdot 5} \\ &= k\sqrt{5} \\ &= 2\sqrt{5} \quad \therefore k = 2 \end{aligned}$$

Then C corresponds to the complex number $2(-1+2i)$
 $= -2 + 4i$

Question 15

$$(a) (i) (\alpha^2 + \beta^2 + \gamma^2) = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\begin{aligned} * \text{ Sum roots } -\frac{b}{a} &= 0^2 - 2 \times 1 \\ \alpha + \beta + \gamma &= -\frac{0}{1} \\ &= 0 \end{aligned}$$

$$\begin{aligned} * \text{ Sum two is } \frac{c}{a} \\ \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\ &= \frac{1}{1} \end{aligned}$$

(ii) From (i), Sum of squares on roots < 0
Then at least one of the roots is complex, since $(z^2) = -1$.

Coefficients are real for $x^3 + x + 12$, so complex roots appear as conjugate pairs. Product of complex conjugates is real, so to get product of roots ($\alpha\beta\gamma = -12$) as real the other root must be real.

$$\text{or let } P(x) = x^3 + x + 12$$

$$P'(x) = 3x^2 + 1$$

Since $x^2 > 0$, $P'(x) \neq 0$, no turning points on this curve and since it is odd degree 3 it only cuts the x -axis at one point. Therefore, only one real root.

(ii) Let complex conjugate roots be $a+bi$ and $a-bi$ [$b \neq 0$]

$$\text{then } x^3 + x + 12 = (x - \alpha) [x - (a+bi)] [x - (a-bi)]$$

$$x^3 + x + 12 = (x - \alpha) [x^2 - 2\alpha x + (\alpha^2 + b^2)]$$

Given $(\alpha^2 + b^2), \alpha = -12$

$$(\alpha^2 + b^2) > 0 \quad \text{so} \quad \alpha < 0$$

With α as the real root, the function will change sign as it passes through α .

Let $\alpha = -3$, $(-3)^3 + (-3) + 12 = -18 < 0$

Let $\alpha = -2$, $(-2)^3 + (-2) + 12 = 2 > 0$

Sign change gives $-3 < \alpha < -2$ *

(b) Let point where the water strikes the wall be (d, h) [h is height above ground]

Given range is $\frac{v^2 \sin 2\theta}{g}$

then $0 < d < \frac{v^2 \sin 2\theta}{g}$

* max distance at $\theta = \frac{\pi}{4}$ is $\frac{v^2}{g}$

Now $0 < \frac{gd}{v^2} < \sin 2\theta$ [$0 < \theta < \frac{\pi}{2}$
 $\Rightarrow 0 < 2\theta < \pi$]

(i) $0 < 2\theta < \frac{\pi}{2}$ ie $0 < \theta < \frac{\pi}{4}$

then $\sin^{-1}\left(\frac{gd}{v^2}\right) < 2\theta$

$\frac{1}{2} \sin^{-1}\left(\frac{gd}{v^2}\right) < \theta$ let $\beta = \frac{1}{2} \sin^{-1}\left(\frac{gd}{v^2}\right)$

gives $\beta < \theta$ [Note: given $V > \sqrt{gd}$
then $0 < \frac{gd}{V^2} < 1$]

(ii) If $\frac{\pi}{2} < 2\theta < \pi$ then $\frac{\pi}{4} < \theta < \frac{\pi}{2}$

and angle $\sin 2\theta = \sin(\pi - 2\theta)$

So $0 < d < \frac{V^2 \sin(\pi - 2\theta)}{g}$

$$\frac{gd}{V^2} < \sin(\pi - 2\theta)$$

$$\sin^{-1}\left(\frac{gd}{V^2}\right) < \pi - 2\theta$$

Given $2\theta < \pi - \sin^{-1}\left(\frac{gd}{V^2}\right)$

$$\theta < \frac{\pi}{2} - \frac{1}{2} \sin^{-1}\left(\frac{gd}{V^2}\right)$$

$$\theta < \frac{\pi}{2} - \beta$$

From (i) & (ii) we have

$$\beta < \theta < \frac{\pi}{2} - \beta$$

(c) let $u^6 = x$

then

$$(u^6)^{\frac{1}{2}} = x^{\frac{1}{2}}$$

$$u^3 = x^{\frac{1}{2}}$$

and

$$(u^6)^{\frac{1}{3}} = x^{\frac{1}{3}}$$

$$u^2 = x^{\frac{1}{3}}$$

on Substitution

$$\int \frac{dx}{x^{\frac{1}{2}} + x^{\frac{1}{3}}} = 6 \int \frac{u^5}{u^3 + u^2} du$$

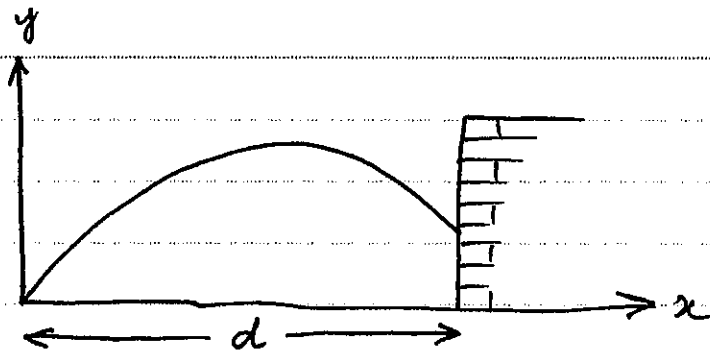
$$= 6 \int \frac{u^3}{u+1} du$$

$$= 6 \int u^3 - u + 1 - \frac{1}{u+1} du$$

also $6u^5 \cdot du = dx$

$$= 6 \left[\frac{1}{4} u^4 - \frac{1}{2} u^2 + u - \ln(u+1) \right] + C$$

$$= 6 \left(\frac{x^{\frac{2}{3}}}{4} - \frac{x^{\frac{1}{3}}}{2} + x^{\frac{1}{6}} - \ln(x^{\frac{1}{6}} + 1) \right) + C$$



$$\ddot{y} = -g \quad \text{--- (1)}$$

$$\dot{y} = -gt + V \sin \theta \quad \text{--- (2)}$$

$$y = -\frac{1}{2}gt^2 + Vt \sin \theta \quad \text{--- (3)}$$

$$\ddot{x} = 0 \quad \text{--- (4)}$$

$$\dot{x} = V \cos \theta \quad \text{--- (5)}$$

$$x = Vt \cos \theta \quad \text{--- (6)}$$

When $x = d$: $Vt \cos \theta = d$

$$t = \frac{d}{V \cos \theta} \quad \text{sub in (3)}$$

$$\therefore y = -\frac{g}{2} \cdot \frac{d^2}{V^2 \cos^2 \theta} + V \sin \theta \cdot \frac{d}{V \cos \theta}$$

$$= -\frac{gd^2}{2V^2 \cos^2 \theta} + d \tan \theta$$

when $x = d$ we want $y > 0$

$$\therefore -\frac{gd^2}{2V^2 \cos^2 \theta} + d \tan \theta > 0$$

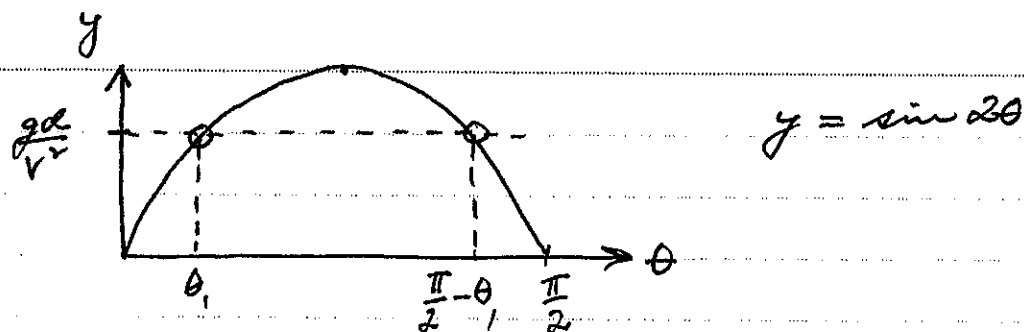
$$-d \left(\frac{gd}{2V^2 \cos^2 \theta} - \tan \theta \right) > 0$$

$$\text{i.e. } \frac{gd}{2V^2 \cos^2 \theta} - \tan \theta < 0$$

$$\Rightarrow \frac{gd}{2V^2 \cos^2 \theta} < \frac{\sin \theta}{\cos \theta}$$

$$x \cos^2 \theta \Rightarrow \frac{gd}{2V^2} < \sin \theta \cos \theta$$

$$\therefore \frac{gd}{V^2} < \sin 2\theta$$



when $\sin 2\theta = \frac{gd}{v^2}$

$$2\theta = \sin^{-1} \frac{gd}{v^2}, \quad (\text{ALSO } \pi - \sin^{-1} \frac{gd}{v^2})$$

$$\Rightarrow \theta = \frac{1}{2} \sin^{-1} \frac{gd}{v^2}, \quad \frac{\pi}{2} - \frac{1}{2} \sin^{-1} \frac{gd}{v^2}$$

Then $\sin 2\theta > \frac{gd}{v^2}$ from above graph

$$\Rightarrow \frac{1}{2} \sin^{-1} \frac{gd}{v^2} < \theta < \frac{\pi}{2} - \frac{1}{2} \sin^{-1} \frac{gd}{v^2}$$

QUESTION 16

(a) (i) $P'(x) = n \cdot x^{n-1} + a$

Given $x = \alpha$ is a double root

then $P(\alpha) = 0 = P'(\alpha)$

So $n \cdot \alpha^{n-1} + a = 0$

$\therefore \alpha^{n-1} = -\frac{a}{n} \quad \dots (I)$

(ii) From (i) $P(\alpha) = 0$

so $\alpha^n + a\alpha - b = 0 \quad \dots (II)$

From (i) $\alpha^n = \left(-\frac{a}{n}\right)\alpha$

So $\alpha \left(-\frac{a}{n}\right) + a\alpha - b = 0$

$a\alpha \left(1 - \frac{1}{n}\right) = b$

$a\alpha = \frac{nb}{(n-1)}$

$\alpha = \frac{nb}{a(n-1)} \quad *$

Sub into II, *

$\left(\frac{nb}{a(n-1)}\right)^n + \left(\frac{nb}{a(n-1)}\right)a - b = 0$

$(nb)^n + nb [a^n(n-1)^{n-1}] - b [a^n(n-1)^n] = 0$

$(nb)^n = a^n b [(n-1)^n - n(n-1)^{n-1}]$

$= a^n b (n-1)^{n-1} [(n-1) - n]$

$n^n b^n = -a^n b (n-1)^{n-1}$

gives $\frac{b^{n-1}}{(n-1)^{n-1}} = -\frac{a^n}{n^n}$

So $\left(\frac{a}{n}\right)^n + \left(\frac{b}{n-1}\right)^{n-1} = 0$

(b) (i) $y = e^{-x} \cdot \sin x$

Now, $e^{-x} > 0$ for all x

Then $y = 0$ when $\sin x = 0$ [$x \geq 0$]

true at $x = \pi, 2\pi, 3\pi, 4\pi, \dots$

So points $P(\pi, 0); Q(2\pi, 0); R(4\pi, 0) \dots$

(ii) Given $\int e^{-x} \cdot \sin x \, dx = -\frac{1}{2} e^{-x} (\sin x + \cos x)$

we have $A_1 = -\frac{1}{2} [e^{-x} (\sin x + \cos x)]_0^\pi$

$$= -\frac{1}{2} [(e^{-\pi} (0 + -1)) - (e^0 (0 + 1))]]$$

$$= \frac{1}{2} (e^{-\pi} + 1)$$

$$A_2 = -\frac{1}{2} [e^{-x} (\sin x + \cos x)]_{2\pi}^{3\pi}$$

$$= -\frac{1}{2} [e^{-3\pi} (0 + -1) - e^{-2\pi} (0 + 1)]$$

$$= \frac{1}{2} (e^{-3\pi} + e^{-2\pi})$$

$$\begin{aligned}
 A_n &= -\frac{1}{2} \left[e^{-x} (\sin x + \cos x) \right]_{(2n-2)\pi}^{(2n-1)\pi} \\
 &= -\frac{1}{2} \left[e^{-(2n-1)\pi} (0+1) - e^{-(2n-2)\pi} (0+1) \right] \neq \\
 &= \frac{1}{2} \left(e^{(1-2n)\pi} + e^{(2-2n)\pi} \right)
 \end{aligned}$$

* Since $\sin(2n-1)\pi = 0 = \sin(2n-2)\pi$
 and $\cos(2n-1)\pi = -1$; $\cos(2n-2)\pi = 1$

We can write $A_n = \frac{1}{2} \left(e^{(1-2n)\pi} + e^{(2-2n)\pi} \right)$

(ii) For G.P. $\frac{A_2}{A_1} = \frac{\frac{1}{2} (e^{-3\pi} + e^{-2\pi})}{\frac{1}{2} (e^{-\pi} + e^0)}$

$$\begin{aligned}
 &= \frac{e^{-2\pi} (e^{-\pi} + 1)}{(e^{-\pi} + 1)} \\
 &= e^{-2\pi}
 \end{aligned}$$

and $\frac{A_3}{A_2} = \frac{\frac{1}{2} e^{-4\pi} (e^{-\pi} + 1)}{\frac{1}{2} e^{-2\pi} (e^{-\pi} + 1)}$

$$= e^{-2\pi}$$

Shown $\frac{A_2}{A_1} = \frac{A_3}{A_2}$ has common ratio $\frac{1}{e^{2\pi}}$

and so it represents a G.P. , $a = \frac{1}{2}(e^{-\pi} + 1)$
 $r = e^{-2\pi}$

Now, $\left| \frac{1}{e^{2\pi}} \right| < 1$

With $|r| < 1$, $S_{\infty} = \frac{\frac{1}{2}(e^{-\pi} + 1)}{1 - e^{-2\pi}}$

$$S_{\infty} = \frac{1+e^{\pi}}{2e^{\pi}} \bigg/ \frac{e^{2\pi}-1}{e^{2\pi}}$$

$$= \frac{(1+e^{\pi})e^{\pi}}{2(e^{2\pi}-1)}$$

$$= \frac{e^{\pi}(1+e^{\pi})}{2(e^{\pi}+1)(e^{\pi}-1)}$$

$$S_{\infty} = \frac{e^{\pi}}{2(e^{\pi}-1)}$$

(iv) Let $B_1, B_2, B_3, \dots$ be areas "below" the curve.

Then $A_1 + A_2 + A_3 + \dots + A_n + |B_1 + B_2 + B_3 + \dots + B_n|$ gives total area bounded by the curve.

$$\int_0^{\infty} e^{-x} \cdot \sin x \, dx = A_1 + A_2 + \dots - (B_1 + B_2 + \dots)$$

$$= \frac{e^{\pi}}{2(e^{\pi}-1)} - (B_1 + B_2 + \dots)$$

From data $\frac{1}{2} = \frac{e^{\pi}}{2(e^{\pi}-1)} - (B_1 + B_2 + \dots)$

Thus $B_1 + B_2 + \dots + B_n = \frac{e^{\pi} - (e^{\pi}-1)}{2(e^{\pi}-1)}$

$$= \frac{1}{2(e^{\pi}-1)}$$

Exact value $\int_0^{\infty} |e^{-x} \cdot \sin x| \, dx = \frac{e^{\pi}}{2(e^{\pi}-1)} + \frac{1}{(e^{\pi}-1)}$

$$= \frac{e^{\pi} + 1}{2(e^{\pi}-1)}$$

QED